

Part IV
Fundamental
Modern Physics

Chapter 22 Energy Quantization

discreteness
the existence of fundamental unit

$$\ell_g = \sqrt{\frac{G\hbar}{c^3}} \approx 1.6 \times 10^{-35} \text{ m}$$

$$e = 1.60217733 \times 10^{-19} \text{ C}$$

On 27th April 1900, British physicist Lord Kelvin (W. Thomson) gave a lecture titled *Nineteenth-Century Clouds over the Dynamical Theory of Heat and Light* to the Royal Institution of Great Britain. In the lecture, he extolled the near completeness of physics. Then, he mentioned:

“In the remote part of sunny sky of physics there are two small puzzling dark clouds.”

"Beauty and clearness of theory... overshadowed by two clouds..."

Michelson Morley experiment
blackbody radiation

22.1 Blackbody radiation

Thermal radiation (infrared-visible-ultraviolet) emitted by ordinary objects depends not only on the temperature, but also on other properties such as

- shape
- surface properties
- material of which it is constructed
- whether or not it reflects the radiation that falls on it from its surroundings

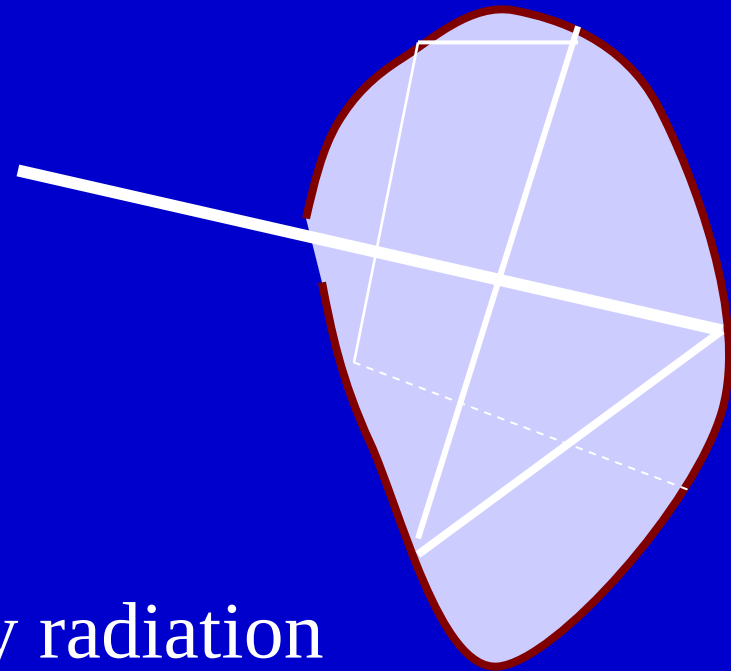
Blackbody a complete black object that reflects none of the radiation that strikes it.

A cavity with a small hole

Radiation from outside that strikes the hole gets lost inside the cavity.

It is the hole, and not the cavity itself, that is the blackbody.

blackbody radiation----cavity radiation



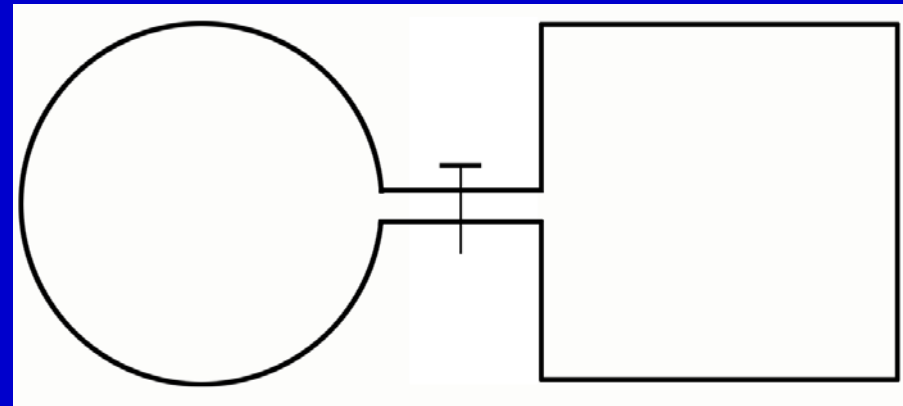
If cavity radiation is in equilibrium state, the energy density u of radiation is

$$u = u(\nu, T) \text{ or } u = u(\lambda, T).$$

If the energy density of the radiation depends on other properties so that

$$u(\nu, T, x) > u(\nu, T, y),$$

a perpetual machine of second kind will become possible.



This is against the second law of thermodynamics.

Therefore, in equilibrium, the energy density of the radiation in a cavity is function of temperature alone.

$$u = u(\nu, T), [u = u(\lambda, T)].$$

the energy density of the radiation in a cavity
per unit range of frequency[wavelength].

$$u(T) = \int_0^{\infty} u(\nu, T) d\nu$$

the energy density of the radiation

$R(\nu)d\nu$ the radiated energy current density of the radiation with frequency in a range $d\nu$

$R(\nu)$ the radiant intensity

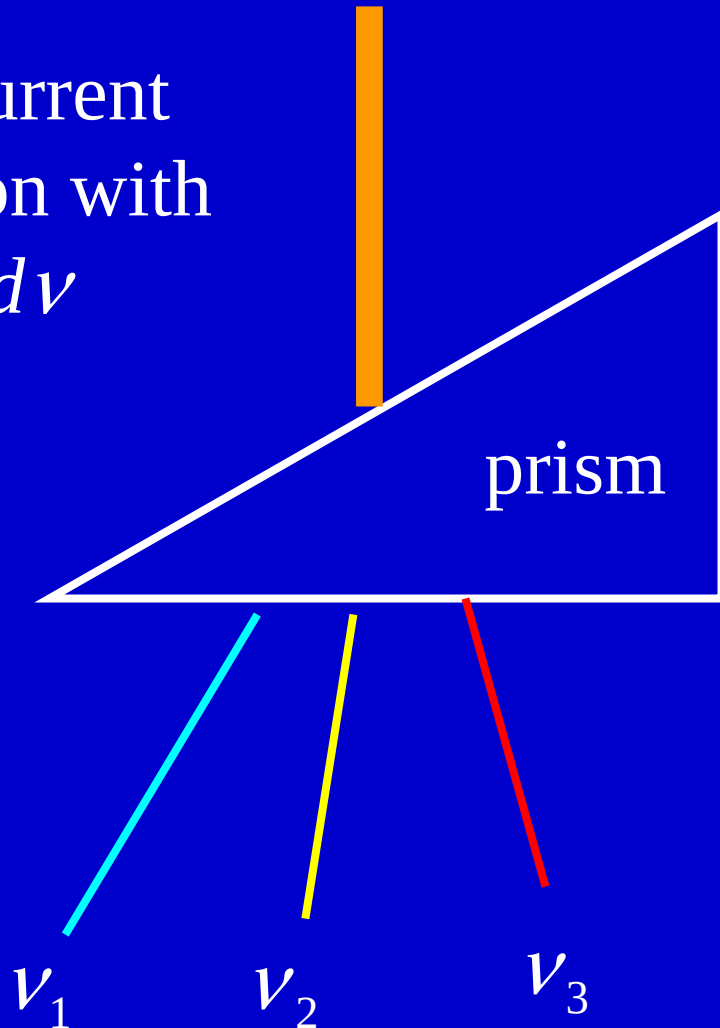
The total radiant intensity

$$\int_0^{\infty} R(\nu) d\nu$$

is just the energy current density j_u .

$$j_u = \frac{1}{4} u c.$$

see effusion § 13.4



The experimental results

1. Stefan-Boltzmann law

$$j_u = \int_0^{\infty} R(\nu, T) d\nu = \sigma T^4.$$

$$\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}.$$



Stefan-Boltzmann's constant

2. The frequency at which the radiant intensity has its peak, which we noted as $\nu_{\max}$ is proportional to the temperature.

$$\frac{h\nu_m}{k_B T} = 2.822 \quad \text{or} \quad \lambda_m T = 2898 \mu\text{m} \cdot \text{K}$$

Wien's displacement law
proposed by Wien in 1893

In 1869 Wien proposed a formula empirically from experimental data:

$$u(\nu, T) d\nu \sim \nu^3 e^{-\alpha\nu/T} d\nu \sim \left(\nu e^{-\alpha\nu/T} \right) \nu^2 d\nu.$$

Wien's formula

Rayleigh-Jeans law

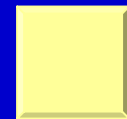
(1900-1905) Lord Rayleigh, Sir James Jeans

1. The cavity is filled with the magnetic standing wave.

2. The number of standing waves with frequencies between ν and $\nu + d\nu$ is

$$N(\nu)d\nu = \frac{8\pi V \nu^2}{c^3} d\nu$$

where V is the volume of the cavity.



3. Each individual standing wave contributes an energy $k_B T$ to the radiation in the cavity. Then, the energy density is

$$u(\nu) d\nu = \frac{8\pi\nu^2}{c^3} \bar{\epsilon} d\nu = \frac{8\pi\nu^2}{c^3} k_B T d\nu.$$



mean kinetic energy

$$\frac{1}{2} k_B T$$

mean potential energy

$$\frac{1}{2} k_B T$$

$$u(T) = \int_0^{\infty} u(\nu, T) d\nu \sim \nu^3 \Big|_0^{\infty} \rightarrow \infty.$$

Ultraviolet catastrophe!!!

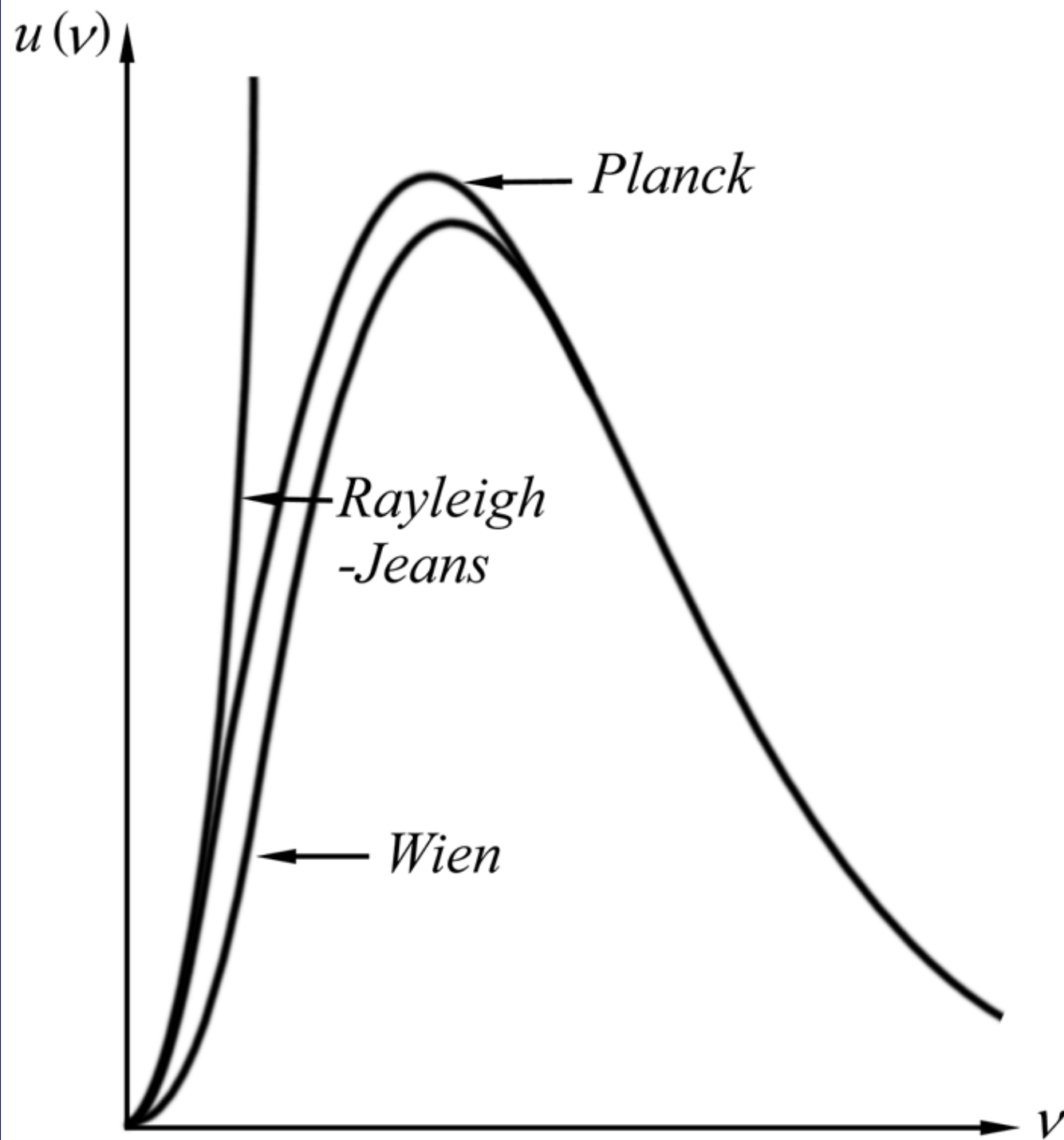
Wien's formula is not in good agreement with experiments at low frequency (long wavelength) limit.

1889 Max Planck proposed a new formula

$$u(\nu, T) d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\beta h\nu} - 1} d\nu$$

Or, equivalently

$$u(\lambda, T) = \frac{8\pi}{\lambda^4} \left(\frac{hc}{\lambda} \right) \frac{1}{e^{\beta hc/\lambda} - 1} d\lambda .$$



$$u(\nu, T) d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\beta h\nu} - 1} d\nu$$

$$\text{high } \nu \text{ limit: } u(\nu, T) d\nu = \frac{8\pi h \nu^3}{c^3} e^{-\beta h\nu} d\nu$$

Wien's formula

$$\text{low } \nu \text{ limit: } \frac{h\nu}{e^{\beta h\nu} - 1} \rightarrow \frac{1}{\beta}$$

$$u(\nu, T) d\nu = \frac{8\pi h \nu^3}{c^3} k_B T d\nu.$$

Rayleigh-Jeans law

Planck interpreted that the oscillators could only emit or absorb energy of **discrete bundles called quanta**:

$$E = nh\nu.$$

$$\bar{\mathcal{E}} = \frac{\sum_n nh\nu e^{-\beta nh\nu}}{\sum_n e^{-\beta nh\nu}} = \frac{h\nu}{e^{-\beta h\nu} - 1}.$$

From

$$\int_0^{\infty} R(\nu, T) d\nu = \sigma T^4,$$

we can obtain

$$\sigma = \frac{2\pi^5 k_B^4}{15c^2 h^3},$$

and

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}.$$

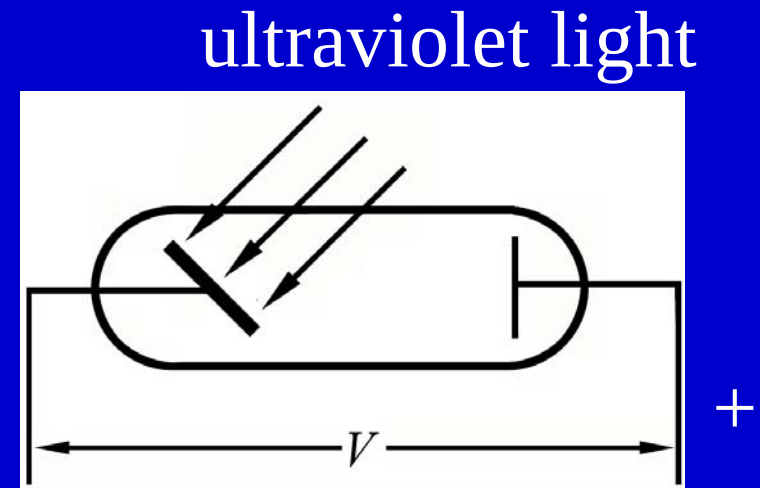
The constant h has fundamental significance. which has no counterpart in classical physics.,

Planck announced the result in 1900. There were no shock waves. Planck himself believed that he had just found a ad hoc explanation.

it was not until after 1911 that Planck fully appreciated the absolutely fundamental nature of quantization

The 1918 Nobel Prize was awarded to Planck for his discovery of energy quanta.

22.2 Photoelectric effect



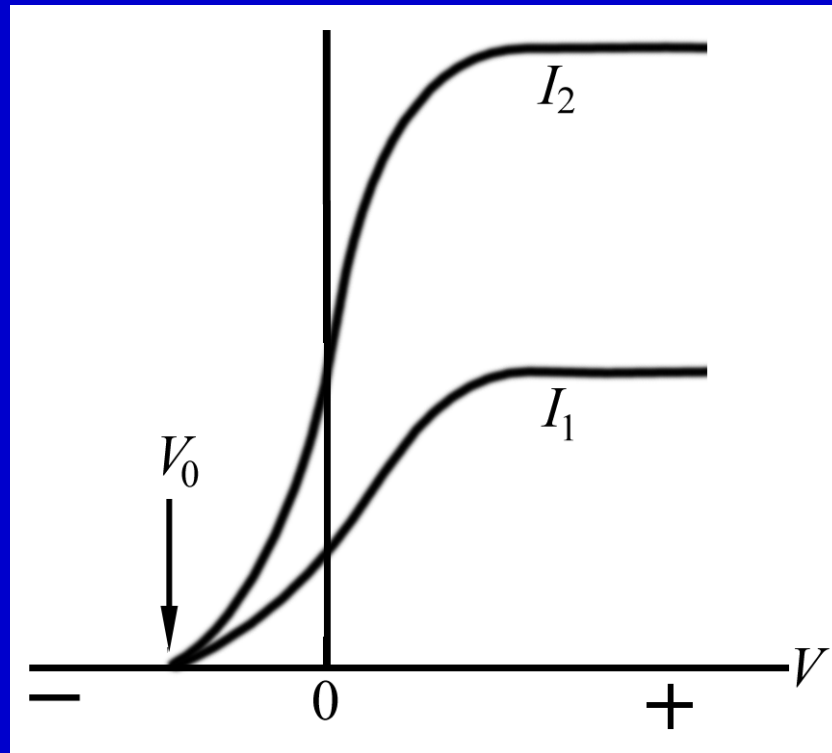
In 1888, Hallwachs found **photoelectric effect**

photo**electrons**

photoelectric current $\sim I$

$$T_{\max} = eV_0 \text{ ? } I_{\text{incident}}$$

stopping potential



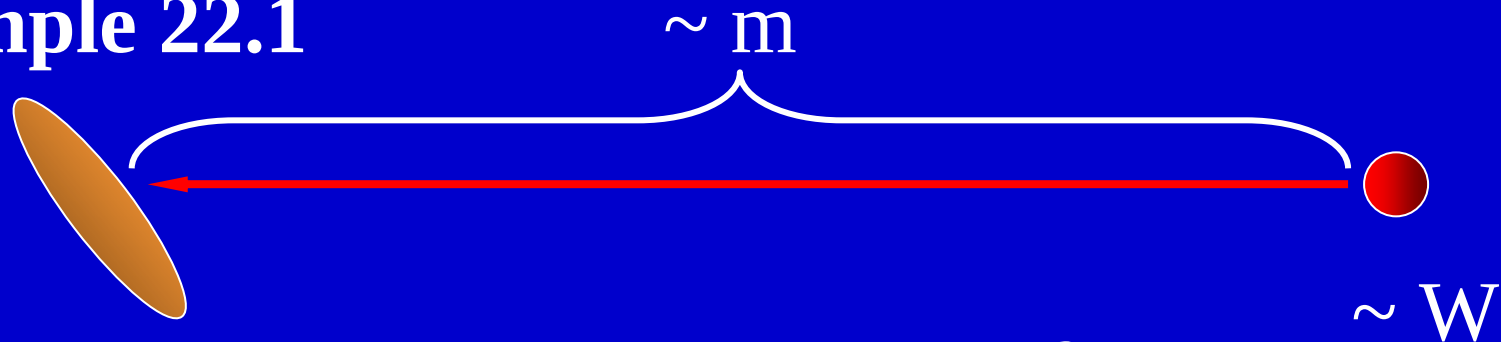
Experiments show

$$T_{\max} \sim \nu$$

ν_C below it, no ...

10^{-9}s after irradiation

Example 22.1



power $P_0 \sim W$; intensity is $P_0 / 4\pi r^2$
absorbed power is

$$P \sim I \pi a^2 \approx \frac{P_0}{4} \left(\frac{a}{r} \right)^2 \sim W (\text{\AA}/\text{m})^2$$

To escape, the electron has to accumulate an amount of energy about eV in a time interval Δt

$$\Delta u \sim \text{eV} \sim W (\text{\AA}/\text{m})^2 \Delta t$$

$$\Delta t \sim \frac{\text{eV}}{W} (\text{\AA}/\text{m})^{-2} \sim 1.6 \times 10^{-19} \times 10^{20} \text{ s} \sim 16 \text{ s}$$

In 1905, Einstein proposed that under some circumstances, light behaves as if its energy was concentrated into localized huddles, **photon**

The energy of a single photon is

$$E = h\nu$$

The maximum kinetic energy of photoelectron is

$$h\nu - \phi = T_{\max}$$

ϕ is called work function

Na	Al	Co	Cu	Zn	Ag	Pt	Pb
2.28	4.08	3.90	4.70	4.31	4.73	6.35	4.14

(eV)

Critical frequency ν_C

$$h\nu - \phi = T_{\max} \geq 0 \quad \nu \geq \frac{\phi}{h} = \nu_C$$

Example 22.2 Critical wavelength and frequency for Na and Pt

green-yellow visible

$$\lambda_C(\text{Na}) = \frac{hc}{\phi} = \frac{1240\text{eV} \cdot \text{nm}}{2.28\text{eV}} = 543\text{ nm}$$

$$\nu_C(\text{Na}) = \frac{2.28\text{eV} \times 3 \times 10^8 \text{ m/s}}{1240 \times 10^{-9} \text{ eV} \cdot \text{m}} = 5.52 \times 10^{14} \text{ Hz}$$

$$\lambda_C(\text{Pt}) = 195\text{ nm} \quad \text{ultraviolet (below 380nm)}$$

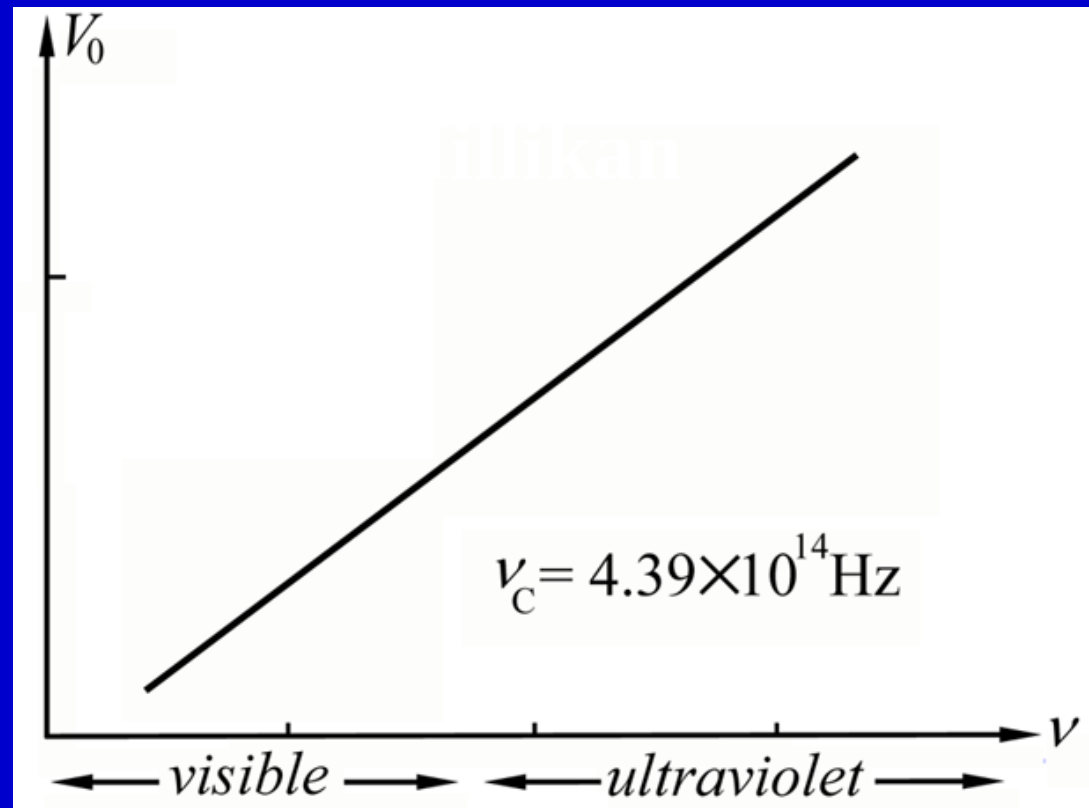
$$\nu_C(\text{Pt}) = 154 \times 10^{14} \text{ Hz}$$

Other results:

$$I_{\text{incident}} = h\nu \times J_{\text{photon}}$$

$$I_{\text{photoelectron}} \sim J_{\text{photon}} \sim I_{\text{incident}}$$

$$V_0 = \frac{h\nu}{e} - \frac{\phi}{e}$$



1923 Nobel prize was awarded to Millikan for his work on the elementary charge of electricity and on the photoelectric effect.

According to photon theory, Planck distribution can be written as

$$u(\nu, T) d\nu = h \nu dn(\nu)$$

$$dn(\nu) = \frac{8\pi \nu^2}{c^3} \frac{1}{e^{\beta h \nu} - 1} d\nu$$

The 1921 Nobel Prize was awarded in 1922 to Einstein for service to theoretical physics, and especially for his discovery of the law of the photoelectric effect.

Assignment: 22.1, 22.4 , 22.5

22.3 Compton effect

Albert Einstein: light quantum (now photon)

The energy of a photon is

$$E = h\nu.$$

The energy-momentum relation:

$$E^2 = m_0^2 c^4 + p^2 c^2.$$

m_0 ——— rest mass

From $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$, we have $v=c \rightarrow m=\infty$.

m is finite. $\longrightarrow m_0=0. \Rightarrow E=pc, \quad p=h(v/c)=h/\lambda.$

reflection

refraction

diffraction

scattering

...

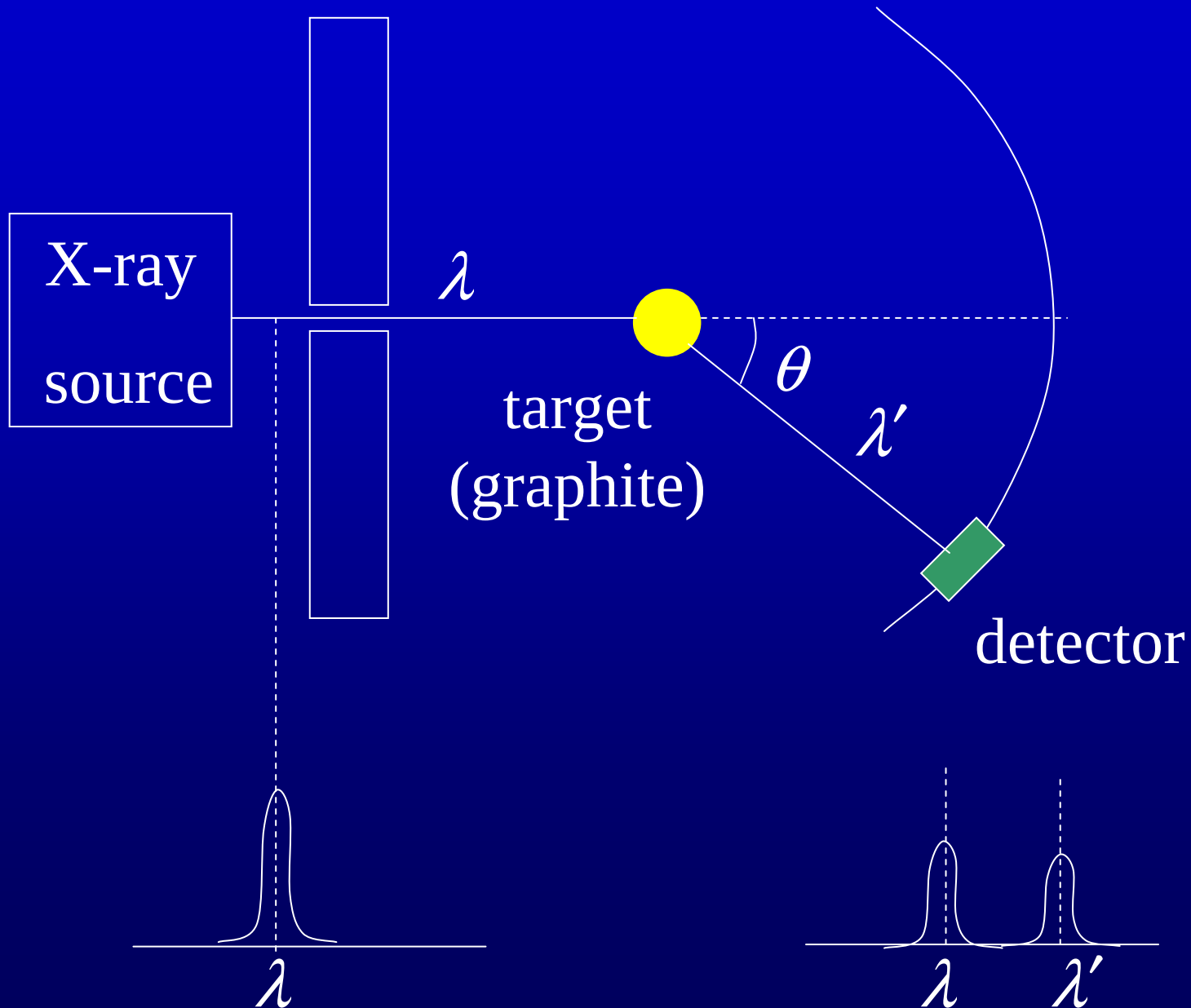
no change in frequency

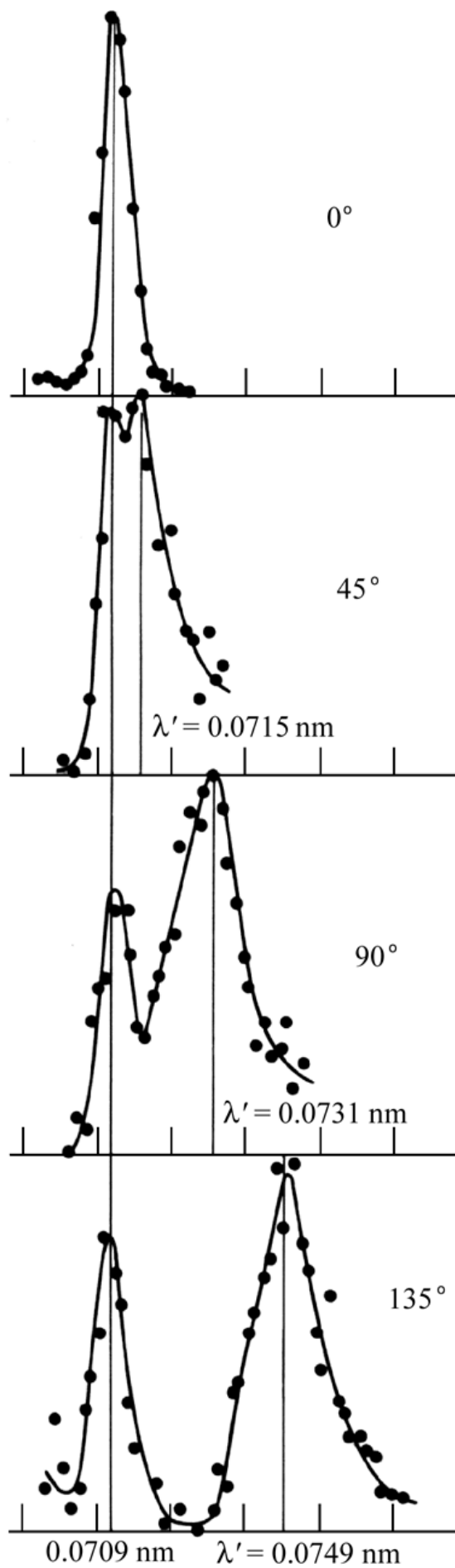
Before 1920, the change in frequency when a beam of X-ray was incident on the matters was observed.

In 1923, Arthur Compton :

The scattering process can be analyzed simply as an interaction(a 'collision') between a single photon and an electron.

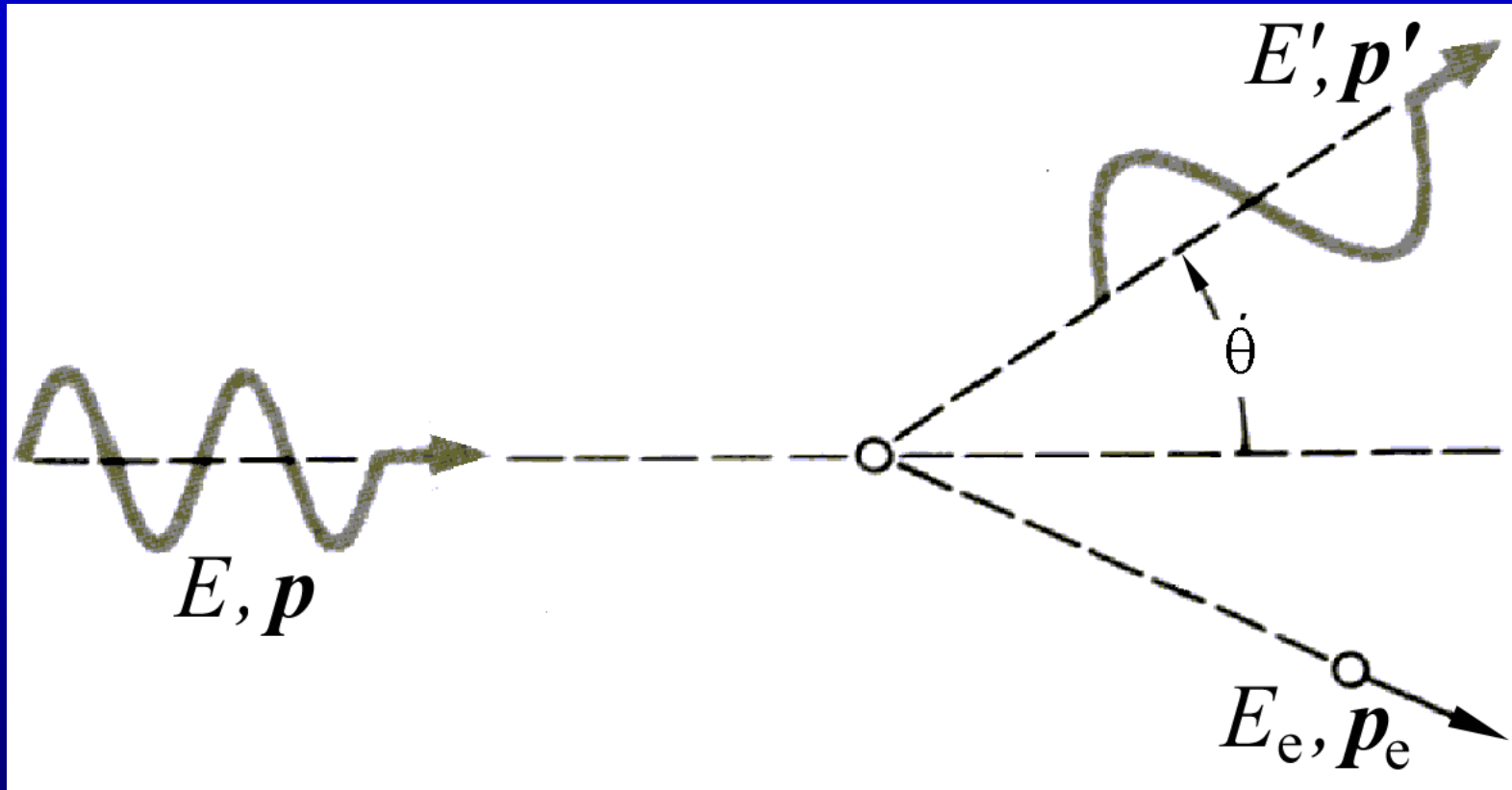
The experimental arrangement of A. Compton





photon

loosely bound, nearly free electron
(assumed to be at rest)



conservation of energy:

$$E + m_0 c^2 = E' + E_e .$$

conservation of momentum:

$$\mathbf{p} = \mathbf{p}' + \mathbf{p}_e$$

energy momentum relation:

$$E_e^2 = c^2 p_e^2 + m_0^2 c^4, \quad E = cp, \quad E' = cp'.$$

$$p_e^2 = |\mathbf{p} - \mathbf{p}'|^2 = p^2 - 2pp' \cos \theta + p'^2$$

$$(E - E' - m_0 c^2)^2 = E_e^2 = c^2 p_e^2 + m_0^2 c^4.$$

$$(cp - cp' + m_0 c^2)^2 = c^2 (p^2 - 2pp' \cos \theta + p'^2) + m_0^2 c^4.$$

$$2(cp - cp')m_0 c^2 - 2c^2 pp' = -2c^2 pp' \cos \theta$$

$$2(cp - cp')m_0 c^2 = 2c^2 pp'(1 - \cos \theta).$$

$$m_0 c \left(\frac{1}{p'} - \frac{1}{p} \right) = 1 - \cos \theta$$

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$\lambda' - \lambda = \left(\frac{h}{m_0 c} \right) (1 - \cos \theta)$$

$$\lambda \equiv \frac{h}{m_0 c} = 2.426 \times 10^{-12} \text{ m}$$

is the Compton wavelength



The peak at 0.070 9nm corresponds to scattering from tightly bound “inner” electrons of the atom.

$$\Delta\lambda = \frac{h}{m_A c} (1 - \cos \theta).$$

Example 22.3. Energy of X-rays of wavelength 0.070 9nm

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.0709 \text{ nm}} = 17.5 \text{ keV}$$

>> work function

Arthur Compton 1927 Nobel laureate

blackbody radiation from infrared to visible

photoelectric effect from visible to ultraviolet

Compton effect from x-ray to gamma ray

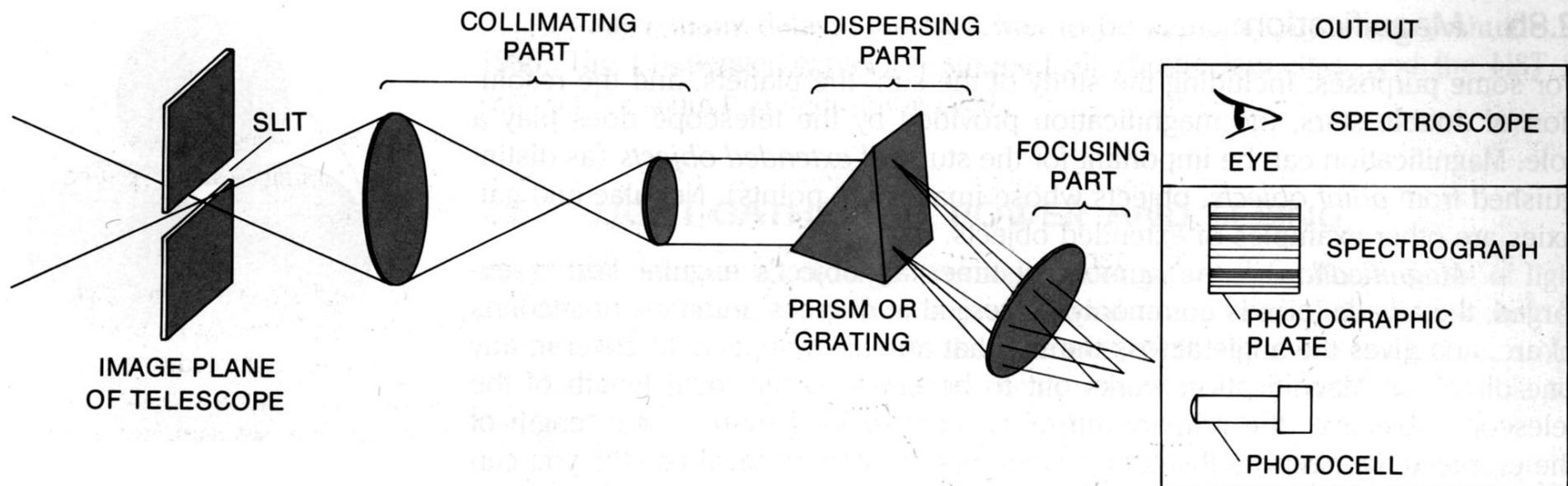
Philippines Bikini  Hawaii

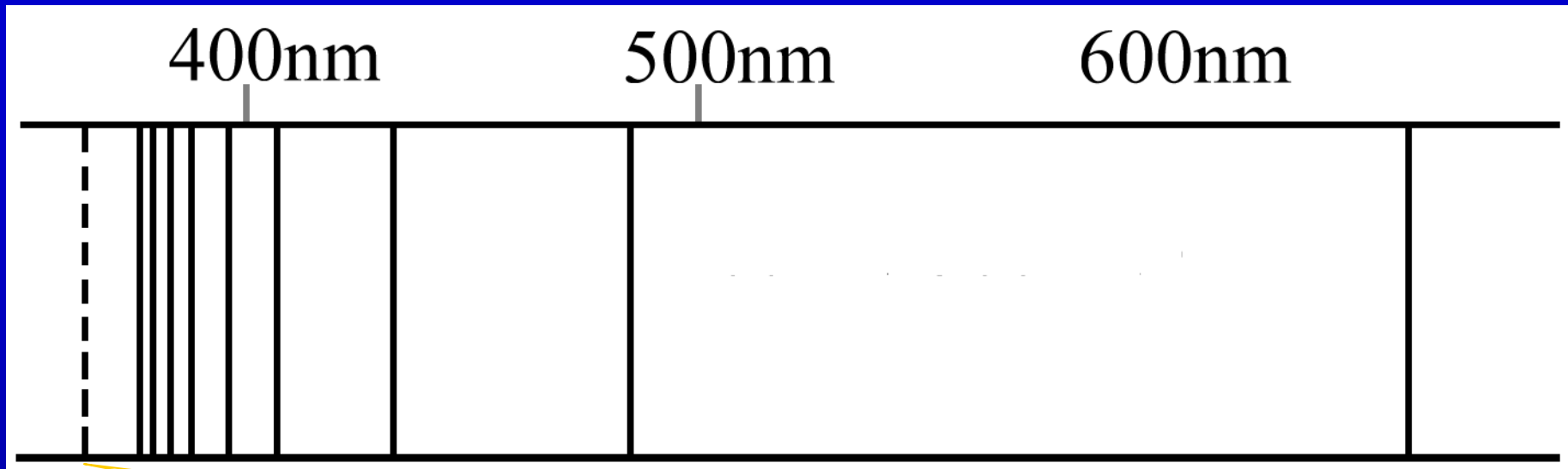
X ray & γ ray  forward surge of electron

Inverse Compton effect: high speed particles transfer energy to space rays.

22.4 Line spectra Energy quantization in atom

Matters in aggregate state, such as filament in a light bulb, or gases under high pressure are heated to incandescence, they emit **continuous spectra** vapors and gases at low pressure : **line emission spectra**





Series limit

In 1885, a Swiss school teacher, Johann Balmer noticed that the wavelengths of the group of emission lines of hydrogen in the visible region could be fitted by the formula

$$\frac{1}{\lambda} = R_{\infty} \left(\frac{1}{2^2} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \dots,$$

Balmer series

where R_∞ is Rydberg constant.

$n=7$ ultraviolet

$n \rightarrow \infty$, series limit $\lambda_{\text{limit}} = \frac{4}{R_\infty} = 364.5 \text{ nm.}$

Rydberg discovered

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \quad \begin{matrix} m = 3, 4, 5, \dots \\ n = m + 1, m + 2, \dots \end{matrix}$$

Lyman series ($m=1$)

Paschen series

($m=3$)

series	discovery	m	n	range
Lyman	1904	1	2,3,4,...	UV (absorb)
Balmer	1885	2	3,4,5,...	visible
Pashen	1908	3	4,5,6,...	infrared
Brackett	1922	4	5,6,7,...	infrared
Pfund	1924	5	6,7,8,...	infrared
Humphreys	1953	6	7,8,9,...	infrared
Hansen & Strong	1973	7	8,9,10,...	infrared

$\alpha, \beta, \gamma, \dots$

A particular strong α line of the Lyman series ($\lambda=121.6\text{nm}$) which could be absorbed by the DNA (deoxyribnucleic acid) is completely absorbed by the atmosphere.

Bohr Model

In 1993, Danish physicist Niels Bohr suggested that atom was like a miniature planetary system.

- Stationary states The electron can stays on 'stationary states' in which the electron has certain values of energy (called energy levels) and would not radiate electromagnetic energy. In these states the orbital angular momentum of the electron takes values that are integer multiples of $\hbar$.

$$mvr = n\hbar \quad (n = 1, 2, \dots).$$

Bohr's quantization condition

- quantum transition The only possible frequencies which result in emission or absorption of radiation are those corresponding to transitions between two energy levels, i.e.,

$$h\nu_{nm} = E_n - E_m .$$

Consider for simplicity the hydrogen atom. The attractive coulomb force provides the centripetal acceleration, so

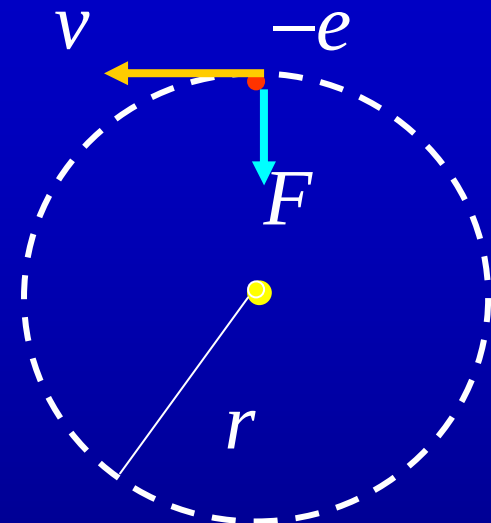
$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r}.$$

Then, the kinetic energy of the electron is

$$E_K = \frac{1}{2}mv^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r}.$$

The potential energy is

$$U = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}.$$



The total energy of the system is thus

$$E = E_K + U = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r}.$$

We use the Bohr's postulate

$$mvr = n\hbar$$

with the expression of the kinetic energy

$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{mvr}{mr}\right)^2 = \frac{1}{2}m\left(\frac{n\hbar}{mr}\right)^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

to obtain a series of allowed values of radius as

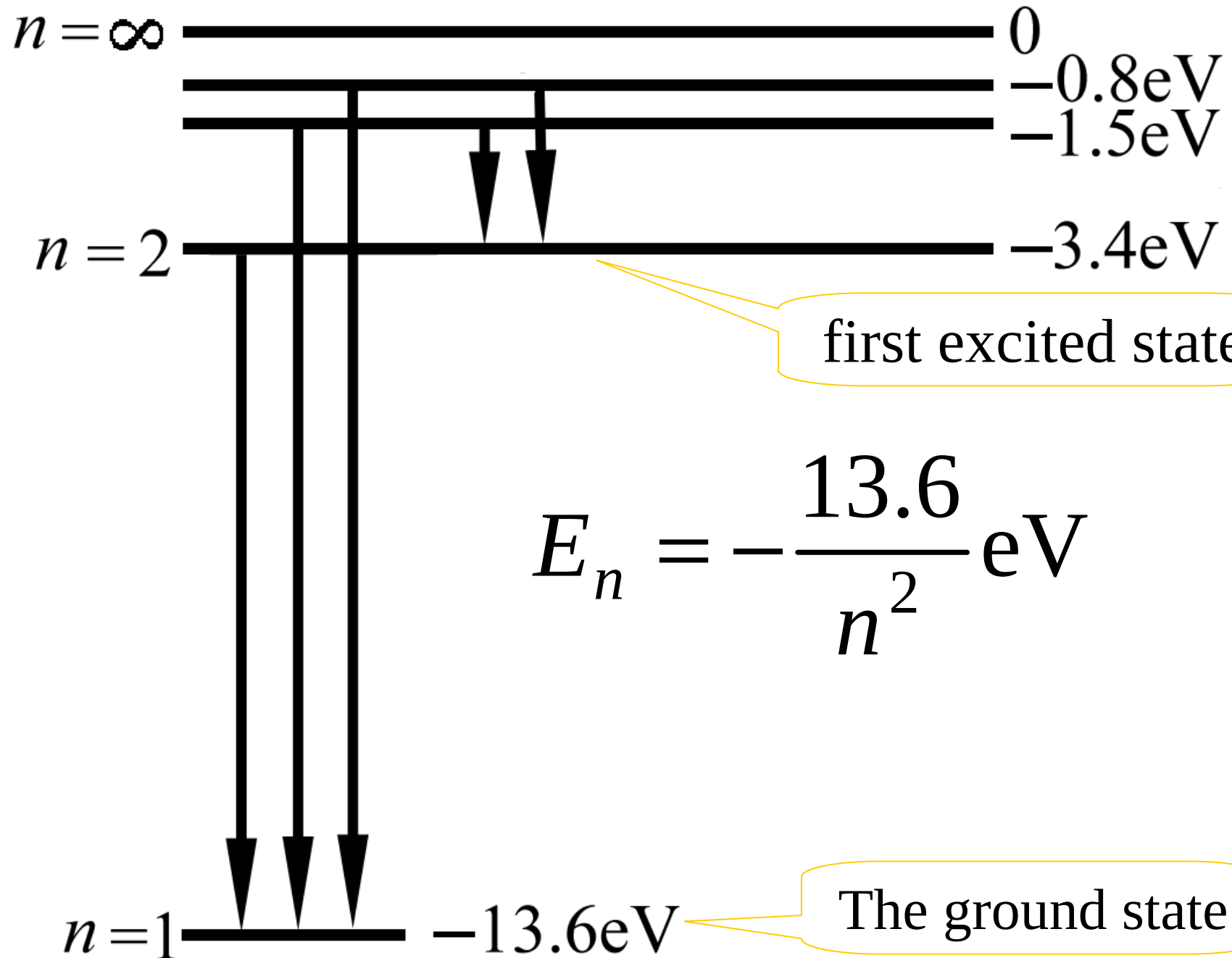
$$r_n = \frac{4\pi\epsilon_0\hbar^2}{me^2} n^2 = a_0 n^2,$$

where the Bohr radius a_0 is defined as

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.0529 \text{ nm}.$$

Therefore, the energy levels are

$$E_n = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{2a_0} \frac{1}{n^2} = -\frac{13.6}{n^2} \text{ eV}.$$



Comparison of Bohr's equation with Rydberg equation yields

$$\begin{aligned} h\nu_{nm} &= hcR_{\infty} \left(\frac{1}{m^2} - \frac{1}{n^2} \right) \\ &= \left(-\frac{hcR_{\infty}}{n^2} \right) - \left(-\frac{hcR_{\infty}}{m^2} \right) = E_n - E_m. \end{aligned}$$

$$\begin{aligned} R_{\infty} &= \frac{1}{4\pi\epsilon_0} \frac{e^2}{2a_0} \frac{1}{hc} = \frac{m_e e^4}{64\epsilon_0^2 \pi^3 \hbar^3 c} \\ &= 10973731.534 \text{ m}^{-1}. \end{aligned}$$

For moving nucleus (of finite mass m')

$$R_m \equiv \frac{\mu}{m_e} R_\infty = \frac{1}{1 + \frac{m_e}{m'}} R_\infty \equiv \frac{1}{\eta} R_\infty ,$$

$$E_n = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{2\eta a_0} \frac{1}{n^2} .$$

For hydrogen-like atoms He^+ , Li^{++} , a factor about Z should be added.

$$E \sim Z^2, \text{ why?}$$

$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{mvr}{mr}\right)^2 = \frac{1}{2}m\left(\frac{n\hbar}{mr}\right)^2 = \frac{1}{8\pi\epsilon_0} \frac{Ze^2}{r} \Rightarrow$$

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{mZe^2} n^2 = \frac{a_0 n^2}{Z},$$

$$E_n = -\frac{1}{8\pi\epsilon_0} \frac{Ze^2}{r_n} = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{2a_0} \frac{Z^2}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}.$$

Rydberg atoms in which an electron has been excited to an unusually high energy level,

diameters reaching 10^{-2} mm, corresponding to a 100 000-fold increase over the diameters of atoms in the ground state

excited states have extremely long lifetimes ~ 1 s
typical lifetimes of lower excited states $\sim 10^{-8}$ s

may be strongly polarized by relatively weak electric fields, or even completely ionized

Rydberg atoms behave therefore in many respects like highly excited hydrogen atoms

In interstellar space, there are atoms n up to 350

In the laboratory, Rydberg atoms with $n \sim 10$ to 290 have been studied.

How good the Bohr model is?

	qualitative	quantitative	
orbit	wrong		uncertainty
$r_0 = a_0 n^2$		wrong	$\langle 1/r \rangle = 1/(n^2 a_0)$
			$\langle r \rangle_{n\ell} = 1/2 [3n^2 - \ell(\ell + 1)] a_0$
			$r_{\text{mp}} (\ell = \ell_{\text{max}} = n - 1) = n^2 a_0$
E_n	correct	correct	$E_n \sim \langle 1/r \rangle_n$
L	correct	wrong	$L < n\hbar$

Assignment: 22.9, 22.12

$$\max(E_e - mc^2) = \frac{E}{1 + \frac{mc^2}{2E}}$$